Gauss quadrature

Gauss 1814, Jacobi 1826

$$\int W(x)f(x)dx = \sum_{i=1}^{i_{max}} w_i f(x_i)$$

if $W(x)$ is a function of a particular kind $(1, e^{-x}, e^{-x^2} \ldots)$, and $f(x)$ is a polynomial of degree $2n-1$ and $i_{max} = n$. The w_i are **weights** that has to be found for each **abscissa** x_i

- Gauss-Legendre $W(x) = 1$

Gauss quadrature
Is it magic?

Example:

$$\begin{aligned}\int_{-1}^{1} f(x)dx &= \int_{-1}^{1}(c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5)dx \\ &= \left[c_0x + \frac{1}{2}c_1x^2 + \frac{1}{3}c_2x^3 + \frac{1}{4}c_3x^4 + \frac{1}{5}c_4x^5 + \frac{1}{6}c_5x^6\right]_{-1}^{1} \\ &= 2c_0 + \frac{2}{3}c_2 + \frac{2}{5}c_4\end{aligned}$$

We need thus to find

$$\sum_{i}^{3} w_i f(x_i) = 2c_0 + \frac{2}{3}c_2 + \frac{2}{5}c_4$$

Guess: Abscissas in $x = 0$ and in $x = \pm a$ Weights: w_0, $w_a = w_{-a}$

$$\sum w_i f(x_i) = \int f(x)dx = 2c_0 + \frac{2}{3}c_2 + \frac{2}{5}c_4$$

$$w_0 f(0) + w_a f(a) + w_a f(-a) = w_0 c_0 + 2w_a(c_0 + c_2 a^2 + c_4 a^4)$$

and we have

$$\begin{aligned} w_0 + 2w_a &= 2 \\ 2w_a a^2 &= \frac{2}{3} \\ 2w_a a^4 &= \frac{2}{5} \end{aligned}$$

$\rightarrow a = \pm\sqrt{3/5},\ w_a = 5/9, w_0 = 8/9$

Listed Abscissas and Weights

always given on the interval -1 to 1

You need to transform to your interval:

$$
\begin{aligned}
&\int_a^b f(x)\,dx = \\
&\left[z = \frac{x-a}{b-a} - \frac{b-x}{b-a} \rightarrow x = z\frac{b-a}{2} + \frac{a+b}{2},\ dx = \frac{b-a}{2}dz\right] \\
&= \frac{(b-a)}{2}\int_{-1}^{1} f(z\frac{b-a}{2} + \frac{a+b}{2})\,dz \\
&\approx \frac{(b-a)}{2}\sum_{i=1}^{n} w_i f(z_i\frac{b-a}{2} + \frac{a+b}{2})
\end{aligned}
$$